Dow Jones Case Study

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Executive Summary

To estimate weekly stock returns for the Dow Jones Industrial Average (DIJA), this study assesses the predictive power of machine learning models, particularly Linear Regression, Support Vector Regression (SVR), and Decision Trees. To evaluate how well these models might account for various forms of market risk during times of volatility, the team integrated basic financial concepts such as the Capital Asset Pricing Model (CAPM).

To compute total weekly returns, the data underwent thorough cleaning and feature engineering, going beyond basic price fluctuations to take the index's entire economic performance into account. To determine each stock baseline for systematic risk (Beta) and determine how susceptible each stock is to more general market fluctuations, we used the CAPM.

A significant performance disparity was found during the "Crash Week" on May 27, 2011, according to the research. All models had trouble understanding the unsystematic risk present in defensive equities like KRFT and MRK, even though the Decision Tree and SVR models offered distinctive insights into non-linear correlations. The defensive stocks' RMSE was higher than that of high-beta industrial stocks like CAT because these stocks saw abrupt declines that departed from their prior systematic tendencies.

Regression and tree-based models are good at capturing systematic market movements, but they are vulnerable to unsystematic events, according to the study's conclusion. This emphasizes the necessity of using hybrid analytical techniques in financial forecasting.

Problem

The underlying volatility of equity markets, which frequently contradicts the predictive assumptions of conventional linear models, is the main obstacle to financial forecasting. The Capital Asset Pricing Model (CAPM) offers a framework for comprehending systematic risk, or the risk connected to a larger market, but it frequently overlooks unsystematic risk, which is made up of distinct or company-specific events that affect individual stocks independent of market averages.

The particular issue of model reliability during times of erratic market behavior is the focus of this study. Defensive equities, like Merck or Kraft, should show steadiness and adhere to their low-beta historical tendencies in normal market situations. The Root Mean Square Error (RMSE) of predictive algorithms such as Linear Regression, SVR, and Decision Trees can be greatly increased by these typically safe stocks experiencing abrupt, non-systematic declines during specific "black swan" events.

Analyzing these various machine learning models' performance in the face of such anomalies is the main goal. This study examines whether industrial, high-beta stocks (like Caterpillar) or stable; defensive stocks are more likely to experience model failure by looking at the "Crash Week" of May 27, 2011. It also investigates whether non-linear techniques like Decision Trees provide any better accuracy than traditional regression when there is unsystematic risk.

Literature Review

Financial forecasting has been widely studied due to its importance in investment decision-making. A fundamental challenge in predicting stock prices is the timing of information

availability. Because financial data is only known at the end of each period, investors must rely on past information to make future decisions. This constraint aligns with the structure of the dataset used in this study, where weekly data is used to predict the following week's returns.

One of the most influential theories in financial economics is the Efficient Market Hypothesis proposed by Eugene Fama. According to Fama (1970), stock prices fully reflect all available information, making it extremely difficult to consistently achieve abnormal returns using historical data. This theory provides an important foundation for understanding why predictive models often struggle to produce accurate forecasts, particularly for short-term price movements.

Despite this limitation, researchers have applied various statistical techniques to identify patterns in financial data. Brown, Pelosi, and Dirska (2013) developed models using variables such as percent change in price, trading volume, and dividend-related metrics to forecast stock returns. Their work demonstrated that while linear regression models can capture some relationships, the overall predictive power remains limited due to weak correlations between predictors and future returns.

To improve predictive performance, more advanced machine learning techniques have been explored. For example, Financial Time Series Forecasting using Support Vector Machines found that Support Vector Machines (SVM) can effectively model nonlinear relationships and often outperform traditional linear models in financial forecasting (Kim, 2003). Similarly, Predicting Stock and Stock Price Index Movement using Machine Learning showed that machine learning methods, including SVM and other algorithms, can provide improved accuracy compared to standard statistical approaches, although results vary depending on the dataset and features used (Patel et al., 2015).

In addition to predictive modeling, risk assessment is a key component of financial analysis. The Capital Asset Pricing Model (CAPM), introduced by William F. Sharpe, measures systematic risk through the beta coefficient (Sharpe, 1964). Beta reflects how sensitive a stock is to overall market movements and is commonly used to classify stocks as high or low risk. Incorporating risk measures alongside predictive models provides a more comprehensive understanding of stock behavior.

Overall, existing literature highlights that while machine learning models can improve predictive accuracy, stock price movements remain difficult to forecast due to weak predictor relationships, market efficiency, and the influence of external factors. These findings guide the selection and evaluation of models in this study.

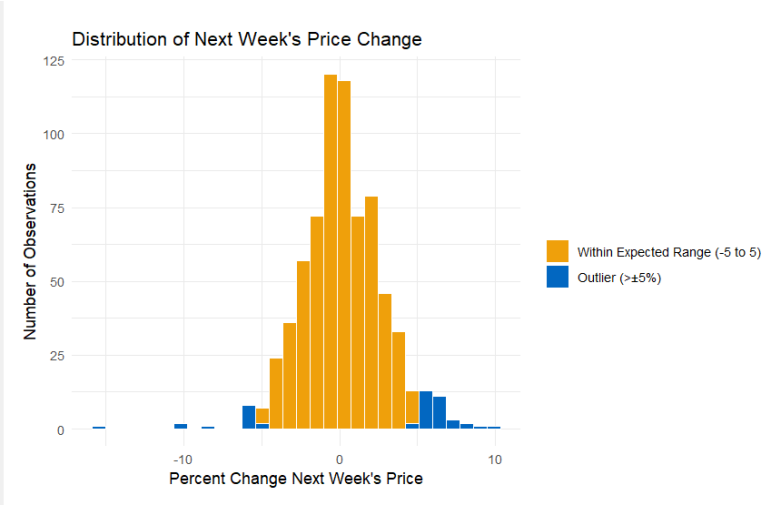
Methods

Data

The study utilizes weekly stock data for 30 Dow Jones companies spanning January through June 2011, sourced from Brown, Pelosi & Dirska (2013). The dataset contains 750 observations and 16 variables, with each row representing one week of trading activity for a single stock. The target variable, *percent_change_next_weeks_price*, captures the percentage return an investor would earn by holding the stock for the following week.

Prior to modeling, we converted price columns containing dollar signs to a numeric format, and removed 30 rows with missing volume data, resulting in 720 clean observations. For each stock, we constructed a lagged variable, *lag_percent_change_price*, to capture the previous week's price change as a potential predictor. Data from Quarter 1 (January–March) was used for

training and Quarter 2 (April–June) for testing, yielding 330 training and 390 test observations for baseline models, and 300 training observations for models incorporating the lagged variable.



Results

Model Performance

The results of this study indicate that predictive performance varied significantly across models and stocks. Model accuracy was evaluated using Root Mean Squared Error (RMSE), with lower values indicating better predictive performance.

Among the three modeling approaches, linear regression produced the weakest results overall. The average RMSE for linear regression was approximately **6.76**, with several stocks showing extremely high error values. For example, CSCO exhibited an RMSE above 50 in the baseline model, indicating very poor predictive accuracy. This suggests that linear models struggle to capture the underlying relationships in the data, which is consistent with the weak correlations observed between predictors and future returns.

In contrast, both decision tree and support vector regression (SVR) models demonstrated substantially improved performance. The decision tree model achieved consistently lower RMSE values across most stocks, typically ranging between 1.4 and 4.5. Similarly, the SVR model produced an average RMSE of approximately **2.91**, outperforming linear regression and showing more stable predictive accuracy across stocks.

Comparison of Model Types

A comparison of model performance shows that nonlinear methods outperformed linear regression in most cases. Both decision trees and SVR were better able to capture patterns in the data, resulting in lower prediction errors across the majority of stocks.

However, the difference between decision tree and SVR performance was relatively small. In some cases, decision trees produced slightly lower RMSE values, while in others, SVR performed better. This suggests that while nonlinear models offer improvements, no single model consistently dominates across all stocks.

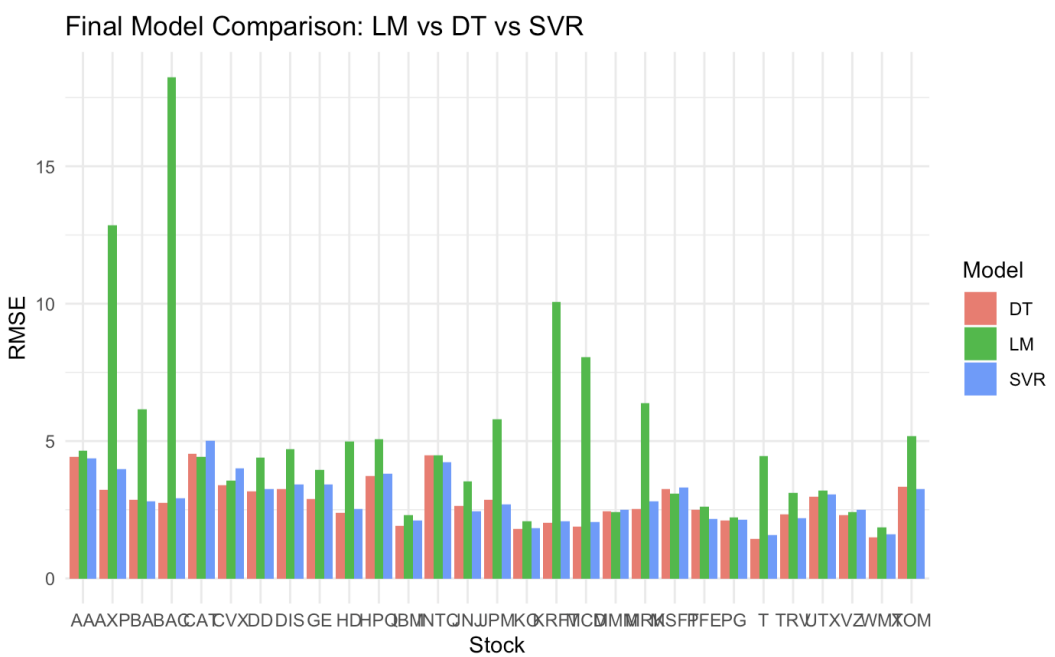


Figure 1: This figure compares RMSE values across all three models. SVR and Decision Trees consistently outperform Linear Regression, indicating that nonlinear methods better capture patterns in stock returns.

Effect of Feature Selection and Lagged Variables

Three model variations were evaluated: a baseline model, a model including lagged variables, and a simplified feature model.

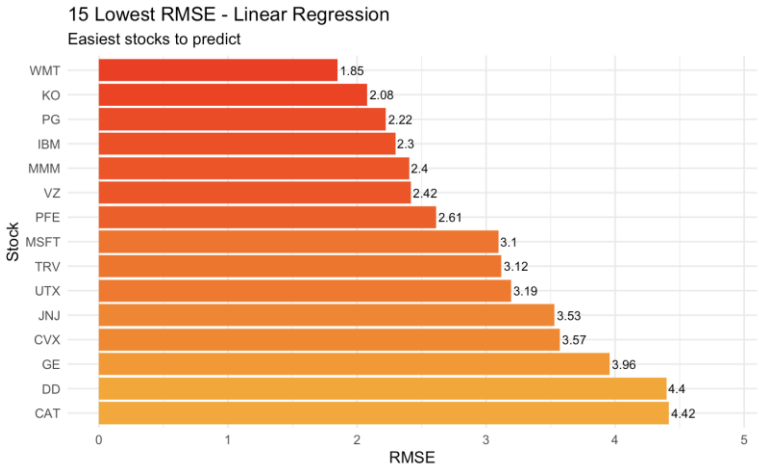
The inclusion of lagged variables (previous week's percent change in price) resulted in only minor and inconsistent changes in RMSE. While some stocks showed slight improvements, others experienced increased prediction error. This indicates that short-term momentum effects are not strong predictors of future weekly returns in this dataset.

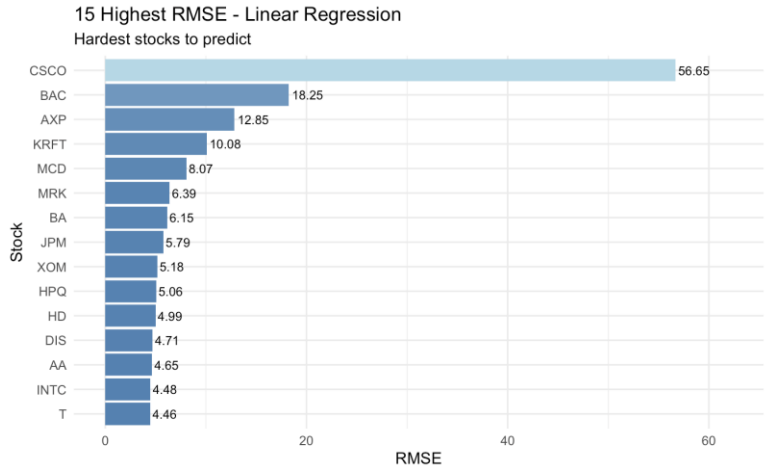
Similarly, simplified models using fewer predictors (volume, closing price, and percent change) did not consistently outperform the baseline models. In some cases, simplified models improved performance, but results were not uniform across all stocks.

Stock-Level Insights

Model performance varied widely across individual stocks. Stocks such as WMT, KO, and PG consistently showed low RMSE values, indicating more stable and predictable behavior. In contrast, stocks like AXP, BAC, and KRFT exhibited higher prediction errors, suggesting greater volatility and unpredictability.

Additionally, the week ending May 27, 2011, showed a market-wide decline across all stocks. This event contributed to increased prediction errors, as models were unable to fully capture this sudden and widespread downturn.





Figures 2 & 3: These figures show variation in predictive difficulty across stocks. Defensive stocks tend to have lower RMSE values, while financial and industrial stocks exhibit higher prediction errors.

Risk Analysis Using CAPM

The CAPM analysis provided further insight into stock behavior by measuring systematic risk through beta values. Stocks with beta greater than 1, such as CAT ($\beta = 1.82$) and BA ($\beta = 1.53$), were classified as high-risk and exhibited greater sensitivity to market movements. In contrast, stocks such as KRFT ($\beta = 0.28$) and PG ($\beta = 0.34$) were categorized as low-risk and demonstrated more stable behavior.

Interestingly, while high-beta stocks are typically expected to experience larger losses during market downturns, the May 27, 2011, event showed deviations from this pattern. Some traditionally low-risk stocks experienced larger declines than expected, highlighting the presence of unsystematic risk.

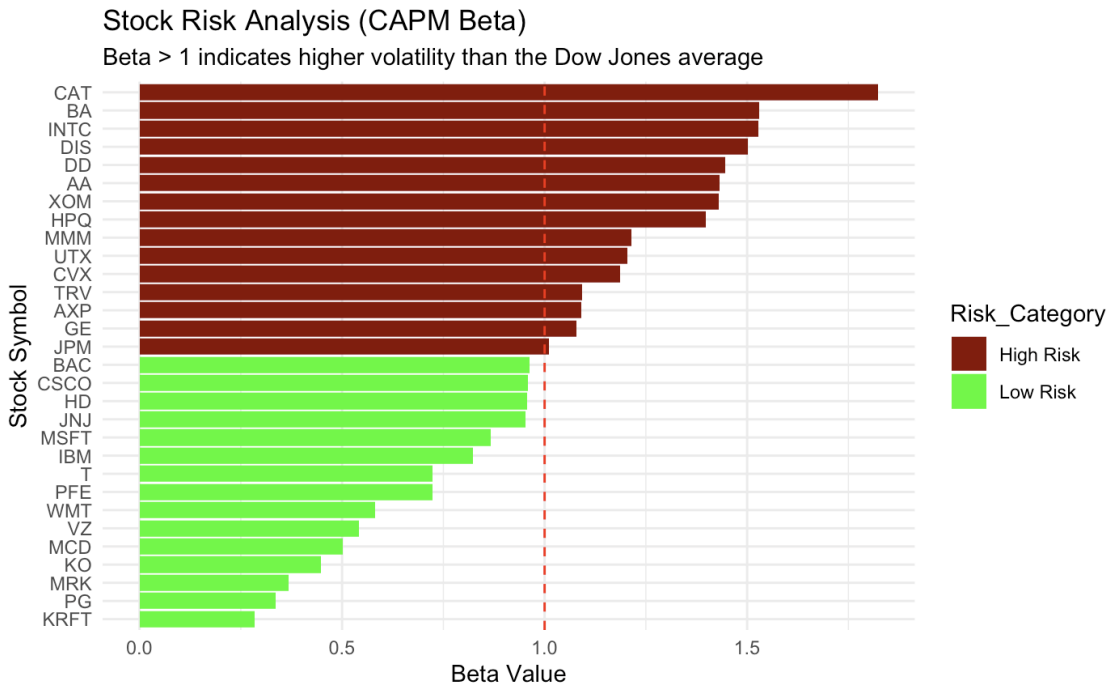


Figure 4: This figure shows beta values across stocks. High-beta stocks are more volatile and respond more strongly to market movements.

Conclusion

In order to assess the predictability of stock returns, this case study effectively combined contemporary machine learning techniques with conventional financial theory. The team went beyond simple price tracking to a thorough examination of total weekly returns by employing Linear Regression, Support Vector Regression, and Decision Trees. The Capital Asset Pricing Model's inclusion made it possible to distinguish between stock-specific unsystematic risk, which frequently makes algorithmic forecasting more difficult, and market-wide systematic risk.

The detection of model failure during “black swan” episodes, particularly the week of May 27, 2011, was the most important finding of this study. The quantitative results showed notable performance gap: traditionally stable defensive stocks saw significant deviations, while high-beta industrial stocks, such as Caterpillar (CAT), which has the highest systematic risk in the index, performed relatively close to their predicted trends. In particular, defensive stocks like Kraft and Merck had a disproportionately high Root Mean Square Error according to the models. This is due to the fact that the models, which were trained on these companies past stability, were unable to explain the abrupt, erratic decline they went through that week.

In the end, these results draw attention to a bsic drawback of quantitative models: their vulnerability to unsystematic risk. The substantial RMSE spikes in defensive sectors indicate that these models need to be complemented with qualitative risk assessments, even though machine learning provides strong capabilities for spotting long-term market trends. The data provides analysts and investors with a textbook example of how “safe” stocks are not immune to volatility and how model accuracy is frequently most jeopardized when market participants shift away from historically stable industries during times of unforeseen uncertainty.

References

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